

CHUANHAN LI

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Education

University of California, Santa Cruz (UCSC)

CA, USA

PhD student @ UCSC CSE

Sep. 2024 - Now

Huazhong University of Science and Technology (HUST)

Wuhan, China

Master of Computer Application Technology

Sep. 2021 – Jul. 2024

Central South University

Changsha, China

Bachelor of Computer Science

Sep. 2017 – Jun. 2021

Research Summary

- A second-year PhD student in computer engineering, systems and hardware security, focusing on extending confidential computing to disaggregated memory and interconnects (CXL).
- First author of a MICRO 2025 paper on efficient security support for CXL memory, built prototype mechanisms on Gem5 and evaluated overheads.
- Strong implementation background: OpenJDK runtime modifications, RDMA optimization, Gem5 simulation, performance analysis on Linux, proficiency in C/C++.

Publications

[MICRO25] **Chuanhan Li**, Jishen Zhao, Yuanchao Xu, “Efficient Security Support for CXL Memory through Adaptive Incremental Offloaded (Re-)Encryption”, the 58th IEEE/ACM International Symposium on Microarchitecture, 2025.

Research Projects

Data Serialization Optimization in RDMA Network

Feb. 2022 - Jun. 2022

- Designed a serialization scheme that preserves in-memory object layout, eliminating pre-transmission copies.
- Cut serialization latency by 42% on microbenchmarks.

ZipGC: Compressed Heap in Managed Runtime atop Disaggregated memory

Feb. 2023 - Feb. 2024

- Extended OpenJDK’s ZGC to identify cold objects and apply base-delta compression.
- Achieved 2.5× bandwidth reduction with <7% runtime overhead on SPECjbb.

AIOR: Efficient Security Support for CXL Memory through Adaptive Incremental Offloaded (Re-)Encryption

Sep. 2024 - Jun. 2025

- Designed adaptive encryption modes switching between AES-XTS and AES-CTR on CXL secure memory.
- Detailed analysis on the pros and cons of AES-XTS and AES-CTR under different memory access behavior characteristics.

DTS: Efficiently Securing CXL Memory via Disaggregated Tiered Security

Mar. 2025 - Now

- Designed adaptive memory encryption with page placement awareness.

Scalable CXL Link-level Integrity and Data Encryption for Multi-host CXL Shared Disaggregated Memory **Sep. 2025 - Now**

- Designed efficient secure CXL Link Data Encryption and Integrity support through cryptographic pad reuse and merge of AES-GCM.

Research Experience

Graduate Researcher - China Grid Computing Laboratory, HUST **Feb. 2021 - Jul. 2024**

- Led two performance improvement studies on a disaggregated memory system with RDMA stack and OpenJDK.
- **Advisor:** Prof. Haikun Liu, Prof. Chencheng Ye

Research Assistant - Center for Research in Systems and Storage, UCSC **Sep. 2024 - Now**

- First author of a MICRO 2025 paper, building secure CXL memory mechanisms and evaluating on representative workloads.
- **Advisor:** Prof. Yuanchao Xu

Skills

Programming Languages: C, C++, Java, Python, Shell

Familiarity: Confidential Computing (AMD SEV/Intel TDX & SGX), AES Encryption Algorithm, RDMA, Linux scripting